CLOUD COMPUTING

Cloud Networking

VPC / VNet — Concepts, Components & Creation
Subnets, CIDR, Gateways & Traffic Flow

Part 1: VPC / VNet

Part 2: Subnets & Gateways



60-Minute Lecture | Cloud Computing Course

Session Agenda

What we'll cover today in 60 minutes

🔵 Part 1 — VPC / VNet Concepts

- What is a Virtual Network? (VPC / VNet)
- Why isolate resources in the cloud?
- VPC Core Components overview
- IP Address Spaces & CIDR Basics
- Creating a VPC — Step by step (AWS & Azure)
- VPC on Azure: Virtual Network (VNet)

🟢 Part 2 — Subnets & Gateways

- Subnets — Segmenting your VPC
- CIDR Notation deep-dive
- Public vs Private Subnets
- Internet Gateway — enabling public access
- NAT Gateway — outbound-only for private resources
- Route Tables — directing traffic
- Security Groups & Network ACLs
- Hands-on: Building a 3-tier VPC

Why Cloud Networking?

The foundation of every cloud deployment

💡 *Analogy: A VPC is like renting your own private floor in a skyscraper (the cloud provider's data centre). You control the layout, keys, and who can enter — but the building infrastructure is managed for you.*

🔒 Isolation

separate your workloads from other tenants. Your resources are invisible to others on the same physical hardware.

🌐 Connectivity

Control exactly which resources can talk to the internet, to each other, and to your on-premises network.

🛡️ Security

Define firewall rules at multiple layers — subnet level and instance level — limiting blast radius of incidents.

⚙️ Control

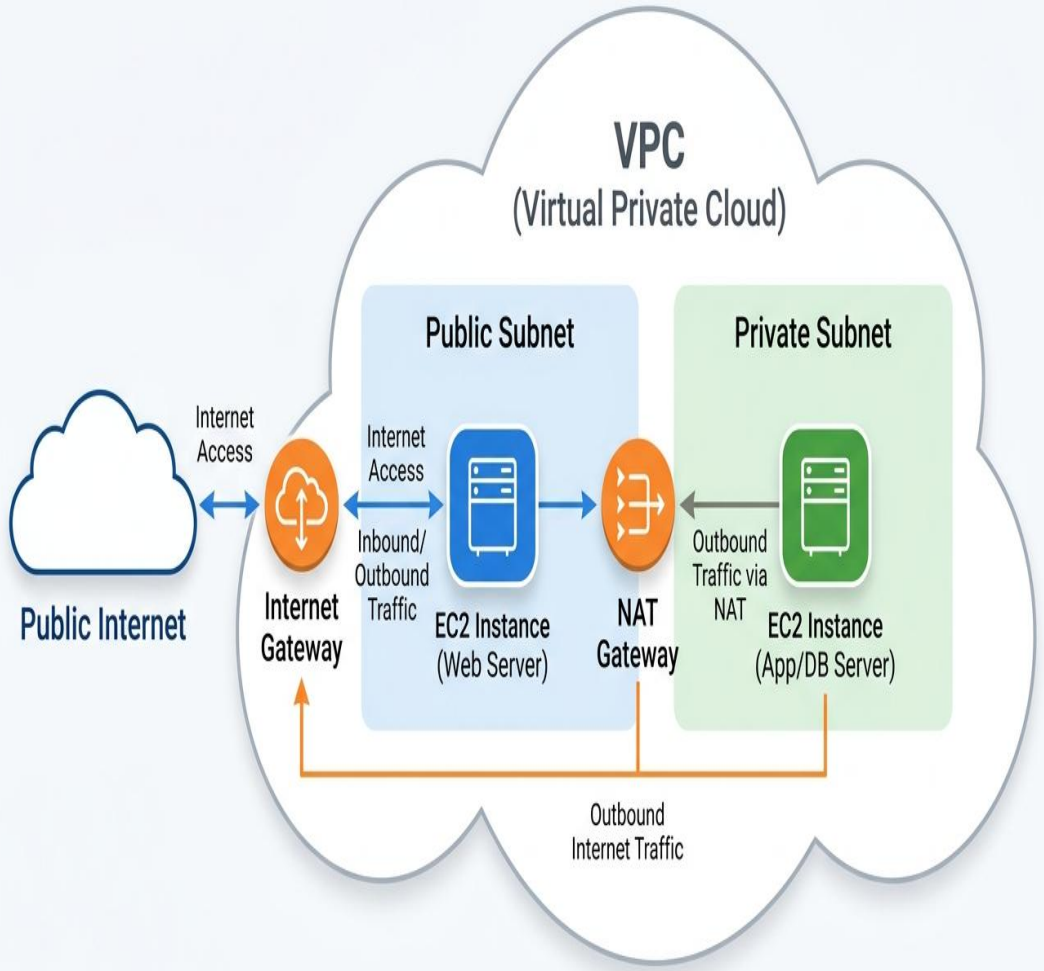
Choose IP ranges, subnets, routing policies, and gateways. Full network topology ownership.

Virtual Private Cloud (VPC) / Virtual Network (VNet)

Part 1 — Core Concept

A VPC (AWS) or VNet (Azure) is a logically isolated section of th cloud provider's network that you define and fully control. It is the networking backbone for all your cloud resources.

- Defined by a private IP address range (CIDR block)
- Spans an entire Region (multiple Availability Zones)
- Resources inside are invisible to the public internet by default
- You control routing, subnets, gateways and security rules
- Each AWS account gets a default VPC; Azure uses VNets per subscription



VPC Core Components

Building blocks you'll configure

CIDR Block

10.0.0.0/16
The IP address range for the entire VPC — typically a private RFC 1918 range.

Subnets

Sub-divisions of the VPC CIDR. Each subnet lives in one Availability Zone.

Internet Gateway

Connects the VPC to the public internet for public-facing resources.

NAT Gateway

Lets private subnet resources initiate outbound internet traffic without being publicly reachable.

Route Table

Defines rules (routes) that determine where network traffic from a subnet is directed.

Security Group / NACL

Virtual firewalls controlling inbound & outbound traffic at instance & subnet level.

AWS VPC vs Azure Virtual Network (VNet)

Two providers, same fundamental concept

AWS — Virtual Private Cloud

- Service name: Amazon VPC
- Scoped to a Region
- CIDR block: /16 to /28
- Default VPC created per region per account
- Uses Security Groups & Network ACLs
- Internet Gateway for public access
- NAT Gateway for private outbound
- Peering, Transit Gateway for inter-VPC

Azure — Virtual Network (VNet)

- Service name: Azure Virtual Network
- Scoped to a Region (tied to subscription)
- CIDR block: /8 to /29 supported
- Must be explicitly created (no default)
- Uses Network Security Groups (NSGs)
- Azure Firewall or NSG for perimeter security
- NAT Gateway or Azure NAT for private outbound
- VNet Peering, Virtual WAN for connectivity

Creating a VPC — Step by Step (AWS)

Hands-on walkthrough

1 Navigate

AWS Console → VPC Dashboard → 'Create VPC'

2 Name & CIDR

Give VPC a name tag. Set IPv4 CIDR: e.g. 10.0.0.0/16

3 Tenancy

Choose 'Default' (shared hardware) or 'Dedicated' tenancy

4 Create

Click 'Create VPC' — takes < 5 seconds to provision

5 Verify

Check VPC list — status shows 'Available'

6 Next steps

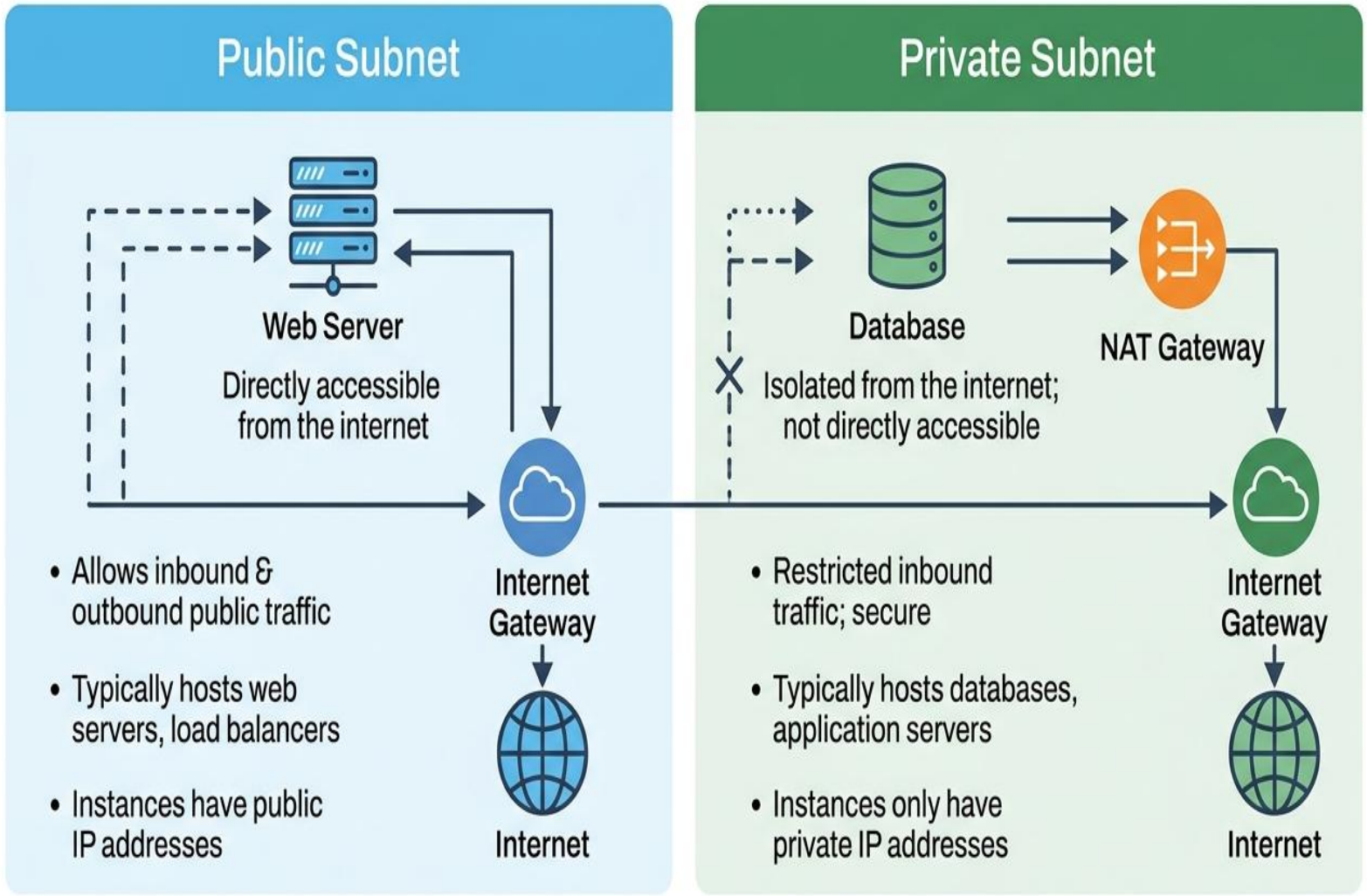
Create Subnets, attach IGW, configure Route Tables

💻 AWS CLI: aws ec2 create-vpc --cidr-block 10.0.0.0/16 --tag-specifications 'ResourceType=vpc,Tags=[{Key=Name,Value=my-vpc}]'

CLOUD NETWORKING

Part 2

Subnets, CIDR Notation, Public vs Private, Gateways & Routing

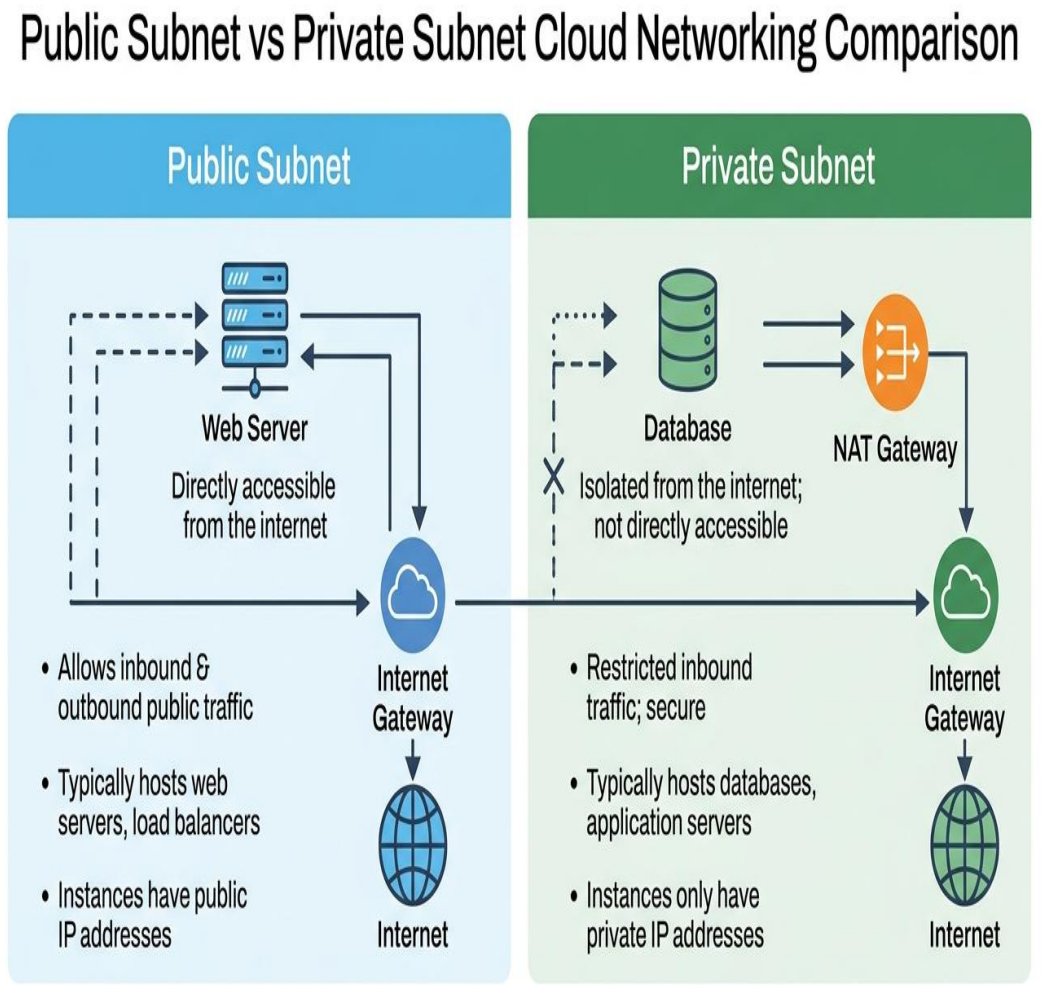


Subnets — Segmenting Your VPC

Dividing your network into smaller, purpose-built pieces

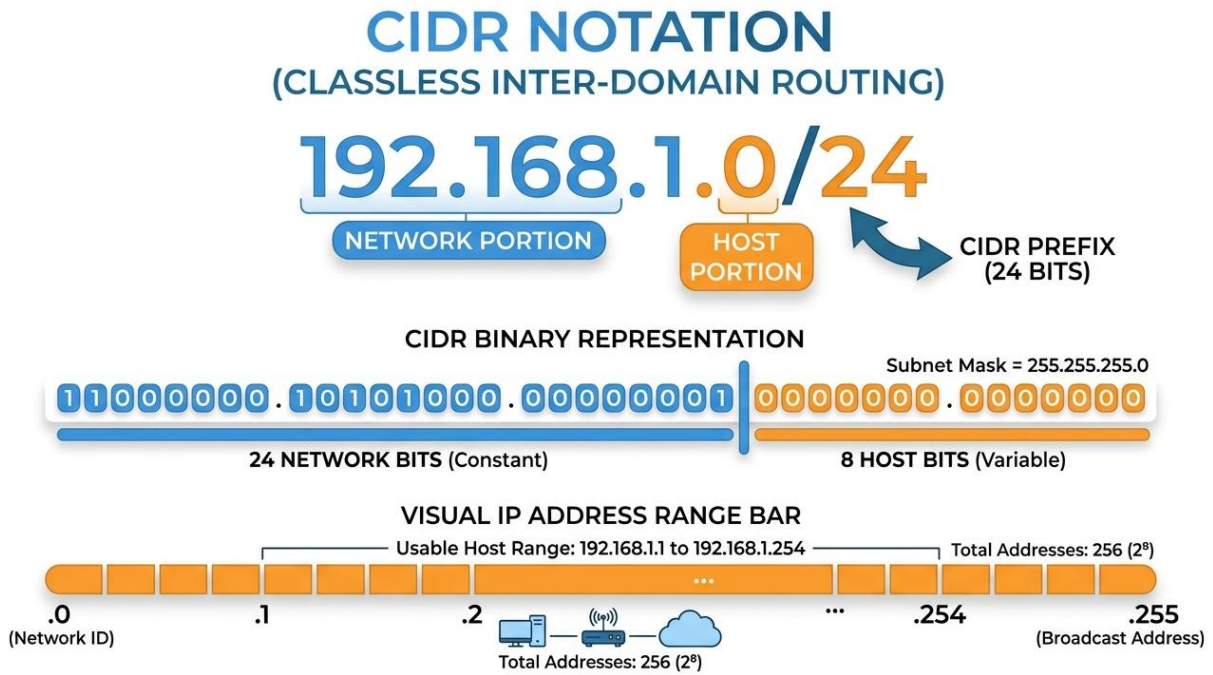
A subnet is a range of IP addresses within your VPC. You assign resources (like EC2 instances) to subnets. Each subnet lives in exactly one Availability Zone.

- A VPC CIDR is divided into smaller subnet CIDRs
- Subnet CIDR must be a subset of the VPC CIDR
- AWS reserves 5 IPs per subnet (first 4 + last 1)
- Subnets in different AZs enable high availability
- A resource in a subnet inherits its routing rules
- Subnets cannot span multiple Availability Zones



CIDR Notation — Classless Inter-Domain Routing

How we define IP address ranges



Reading CIDR: 10.0.0.0 / 16

- 10.0.0.0 → Network Address (base IP)
- /16 → 16 bits are the network prefix
- Remaining 16 bits = host addresses
- 2^{16} = 65,536 total IPs in this block

Private RFC 1918 Ranges: 10.0.0.0/8 | 172.16.0.0/12 | 192.168.0.0/16

CIDR	/8	/16	/24	/26	/28
# IPs	16,777,216	65,536	256	64	16

Public vs Private Subnets

The key design decision in every VPC

🌐 Public Subnet

- Has a route to the Internet Gateway
- Resources can receive public IP addresses
- Accessible from / to the internet
- Hosts: Load Balancers, Web Servers, Bastion Hosts
- Example CIDR: 10.0.1.0/24

⚠️ *Direct internet exposure — secure carefully!*

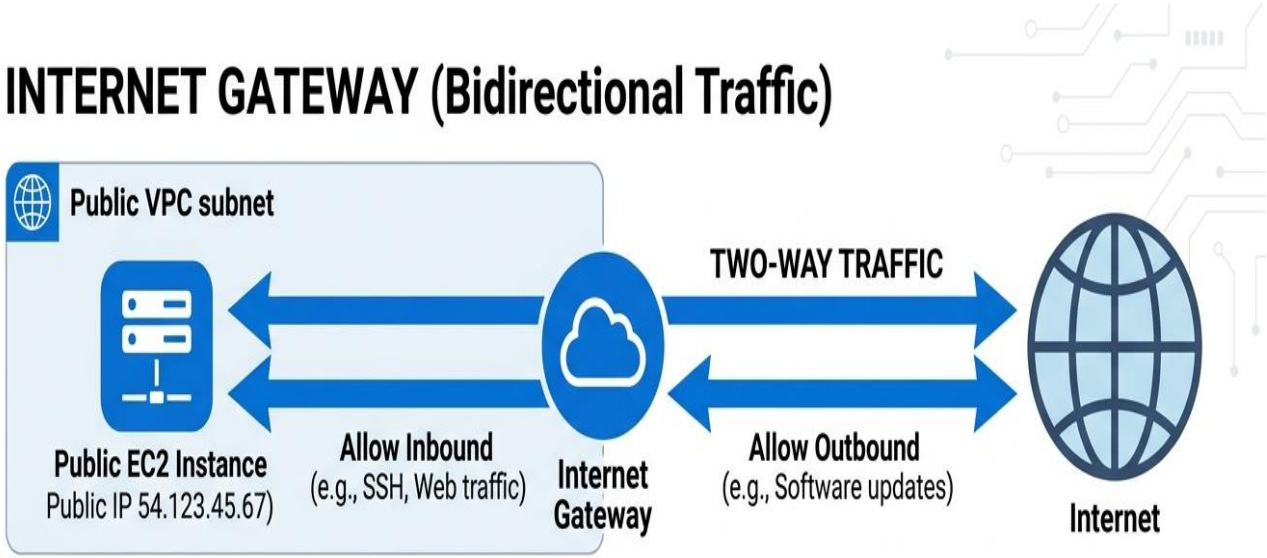
🔒 Private Subnet

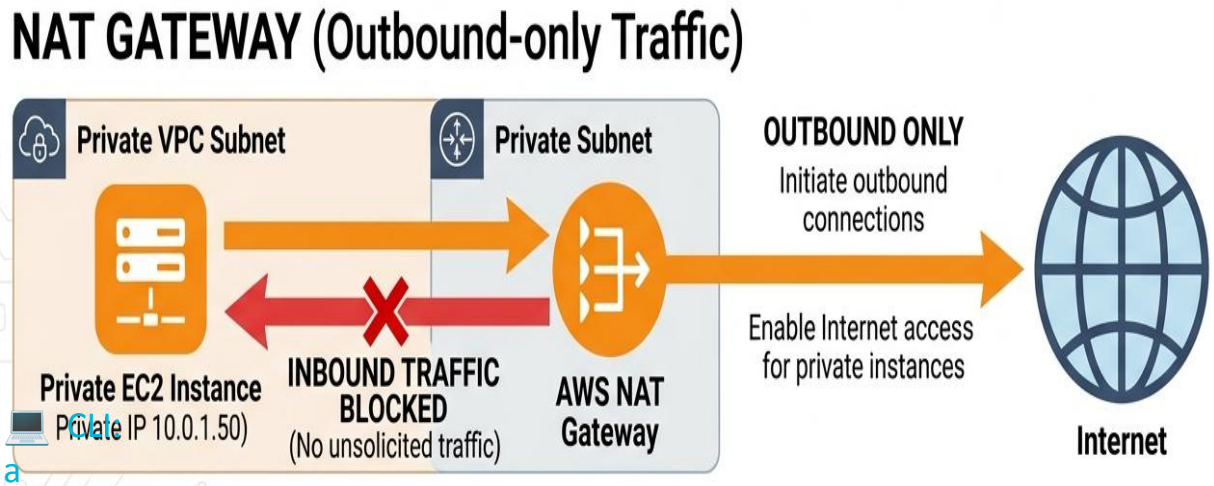
- No direct route to the Internet Gateway
- Resources have NO public IP by default
- Outbound internet via NAT Gateway only
- Hosts: Databases, App Servers, Internal Services
- Example CIDR: 10.0.2.0/24

✅ *Isolated from internet — best for sensitive data*

Internet Gateway (IGW)

The door between your VPC and the internet



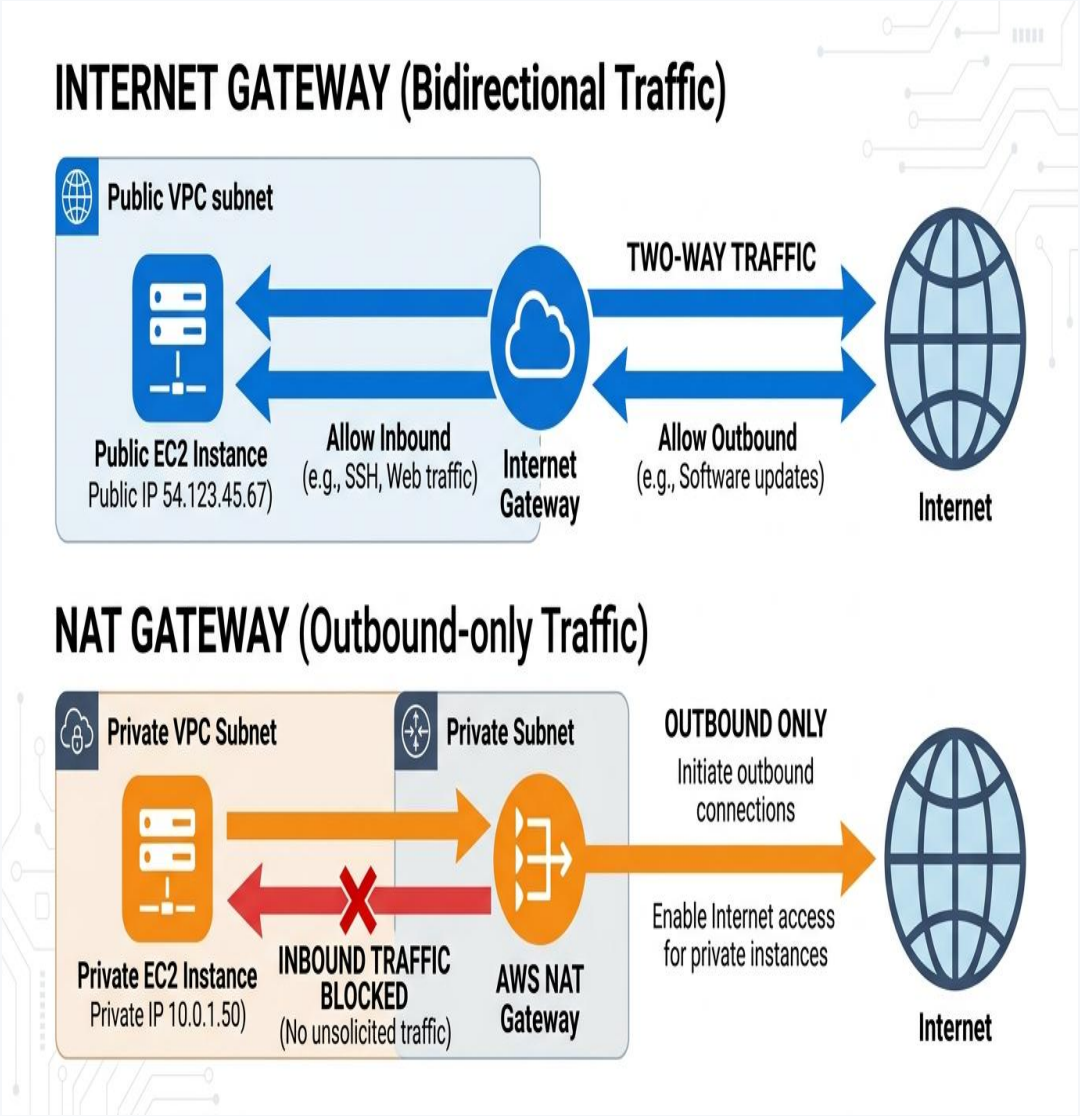


- Attached to a VPC (one IGW per VPC)
- Horizontally scalable, redundant & highly available — no bandwidth limits
- Supports bidirectional traffic (inbound & outbound)
- Resources in public subnets get Elastic IPs or auto-assigned public IPs
- Route table must have 0.0.0.0/0 → IGW for public access
- Free to create; only pay for data transfer

NAT Gateway — Network Address Translation

Outbound internet for private resources

- Placed in a PUBLIC subnet (needs internet connectivity)
- Private subnet instances use NAT to reach internet (OS updates, APIs etc.)
- Translates private IP → public Elastic IP for outbound requests
- Response traffic is allowed back (it's stateful translation)
- Inbound connections from internet are BLOCKED — private resources stay hidden
- Managed service: AWS scales it automatically
- Costs: Hourly charge + data processing charge per GB
- One NAT Gateway per AZ recommended for fault tolerance

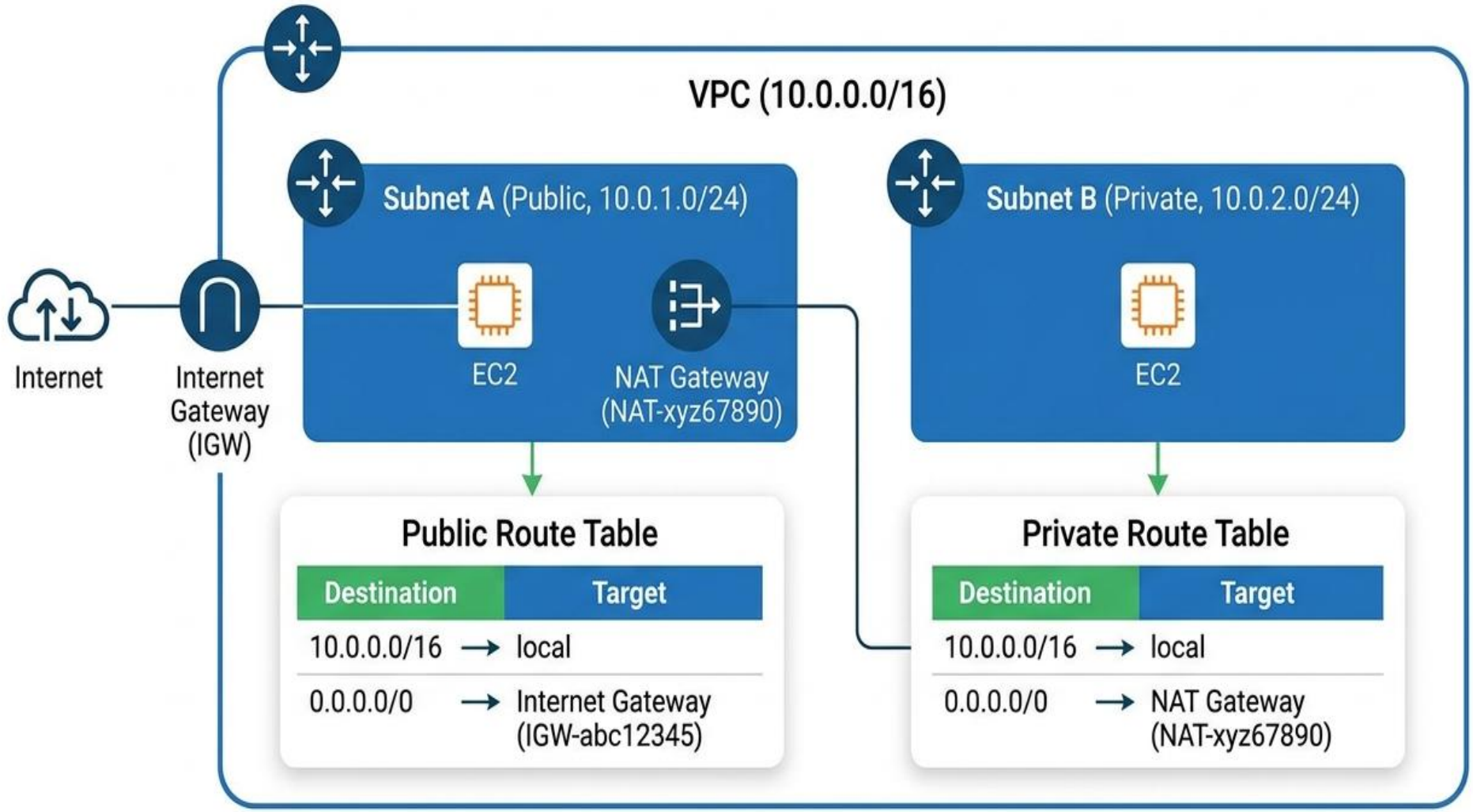


Internet Gateway vs NAT Gateway — Quick Comparison

Choosing the right gateway for your subnet

Feature	Internet Gateway (IGW)	NAT Gateway
Direction	Bidirectional (in + out)	Outbound only (private → internet)
Subnet type	Public subnet	Lives in public; serves private subnets
Public IP?	Yes — instance gets public IP	Yes — NAT Gateway has Elastic IP
Scalability	Unlimited (fully managed)	Scalable, but one per AZ recommended
Cost	Free (pay data transfer only)	Hourly fee + per-GB data processing
Use case	Web servers, load balancers	Databases, app servers needing outbound
Blocks inbound?	No — configure SG/NACL rules	Yes — internet cannot initiate connections

ROUTE TABLES IN CLOUD NETWORKING



Route Tables — Directing Traffic

The GPS of your VPC

Every subnet is associated with a route table that contains rules (routes) specifying where to send traffic destined for specific IP ranges.

Public Subnet Route Table

Destination	Target
10.0.0.0/16	local
0.0.0.0/0	igw-xxxxxxxx

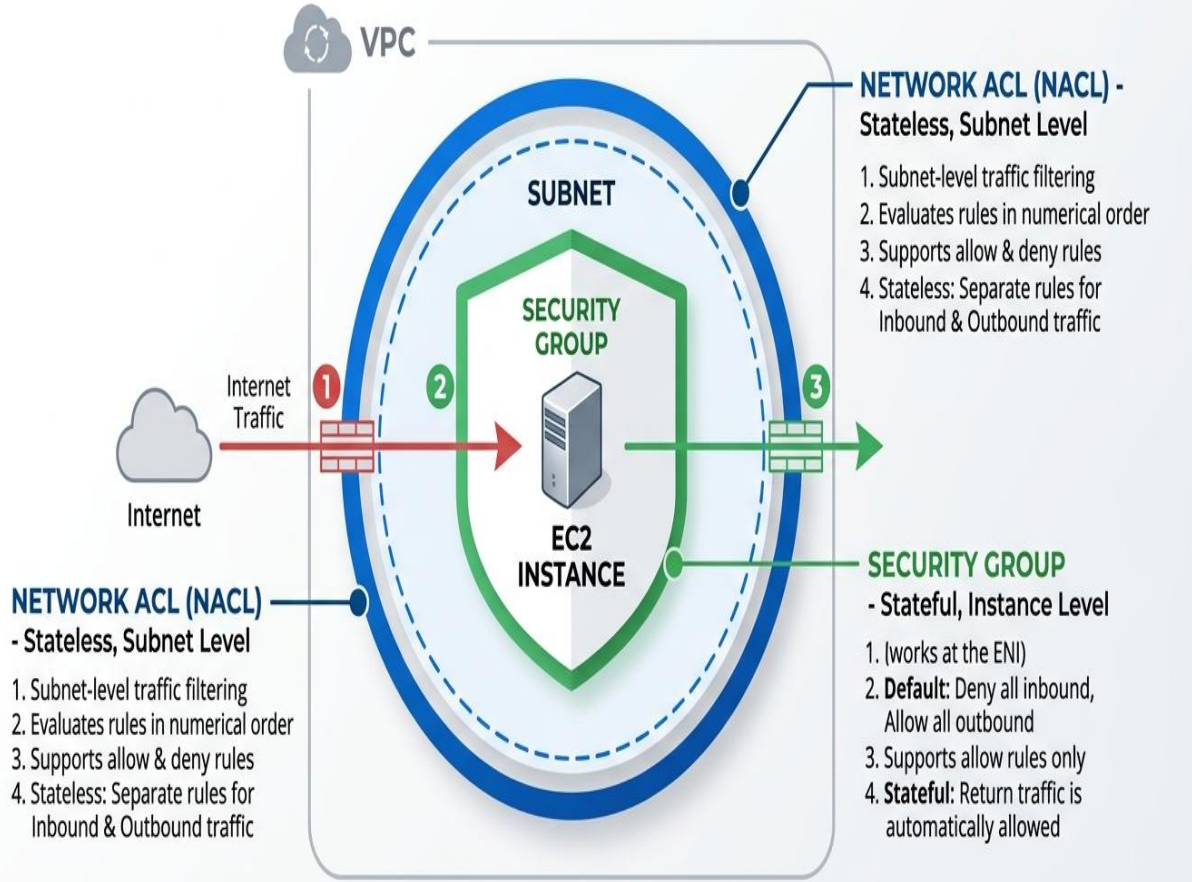
Private Subnet Route Table

Destination	Target
10.0.0.0/16	local
0.0.0.0/0	nat-xxxxxxxx

💡 *Traffic hits the most specific matching route first (longest prefix match)*

Security Groups & Network ACLs

Two layers of traffic filtering



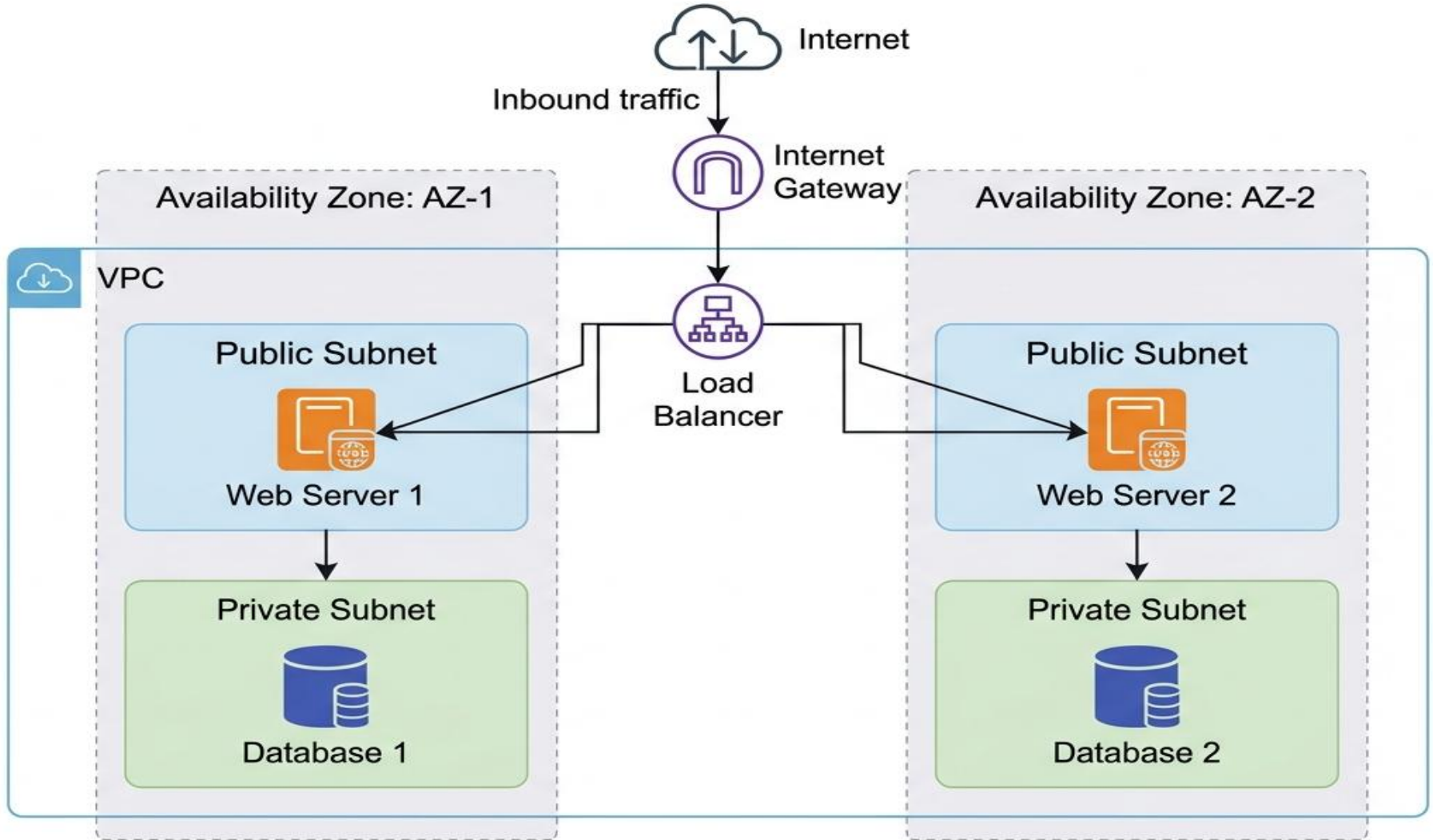
Security Groups (SG)

- Applied at the EC2 instance (ENI) level
- Stateful — return traffic auto-allowed
- Allow rules only (no explicit deny)
- Evaluated as a whole (all rules check)

Network ACLs (NACLs)

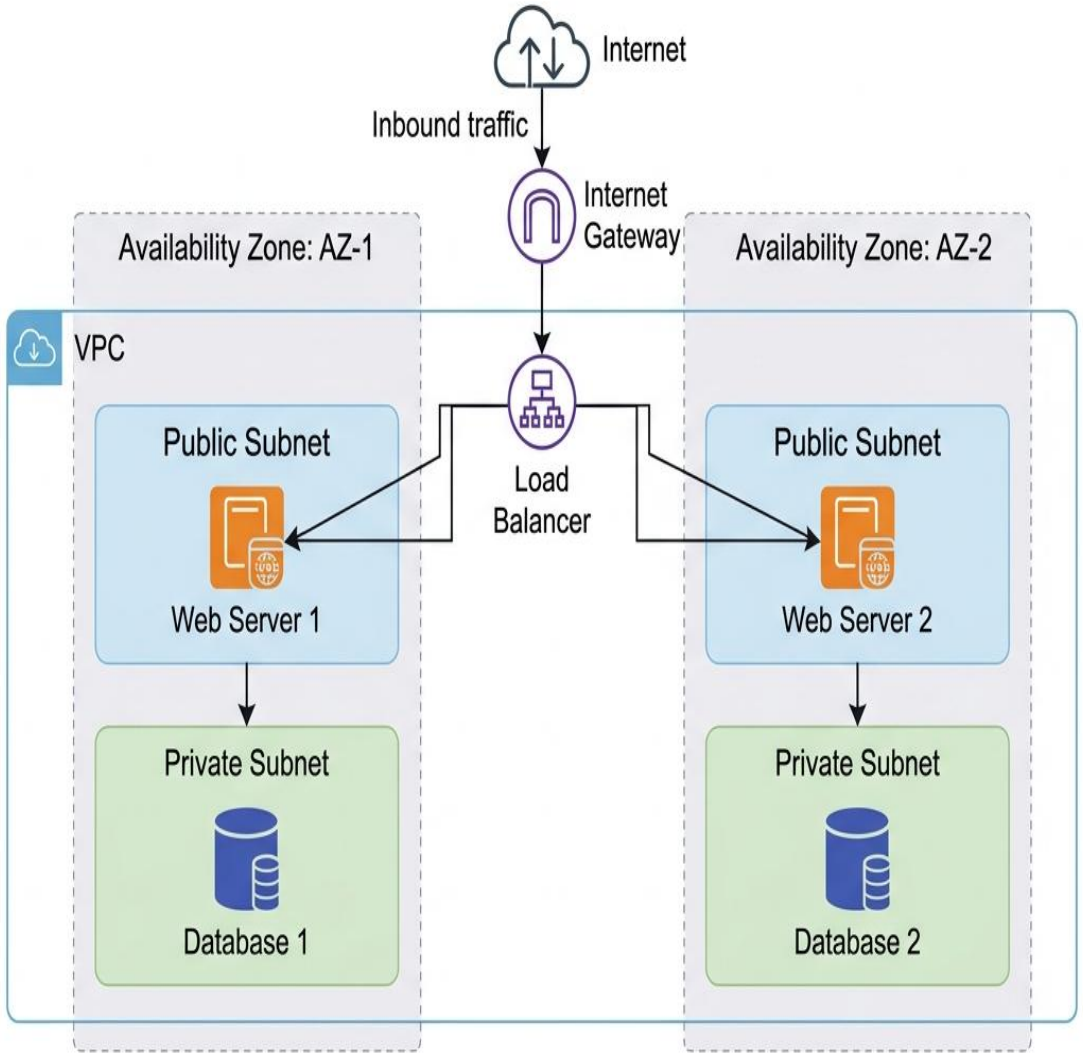
- Applied at the subnet boundary
- Stateless — return traffic must be explicitly allowed
- Supports Allow AND Deny rules
- Rules evaluated in number order (lowest first)

💡 *Best practice: Use Security Groups as primary defense, NACLs as an additional layer for subnet-level blocking.*


Internet
Inbound traffic
Internet Gateway
Availability Zone: AZ-1
Availability Zone: AZ-2
VPC
Load Balancer
Public Subnet
Web Server 1
Public Subnet
Web Server 2
Private Subnet
Database 1
Private Subnet
Database 2

Complete VPC Architecture — Multi-AZ Design

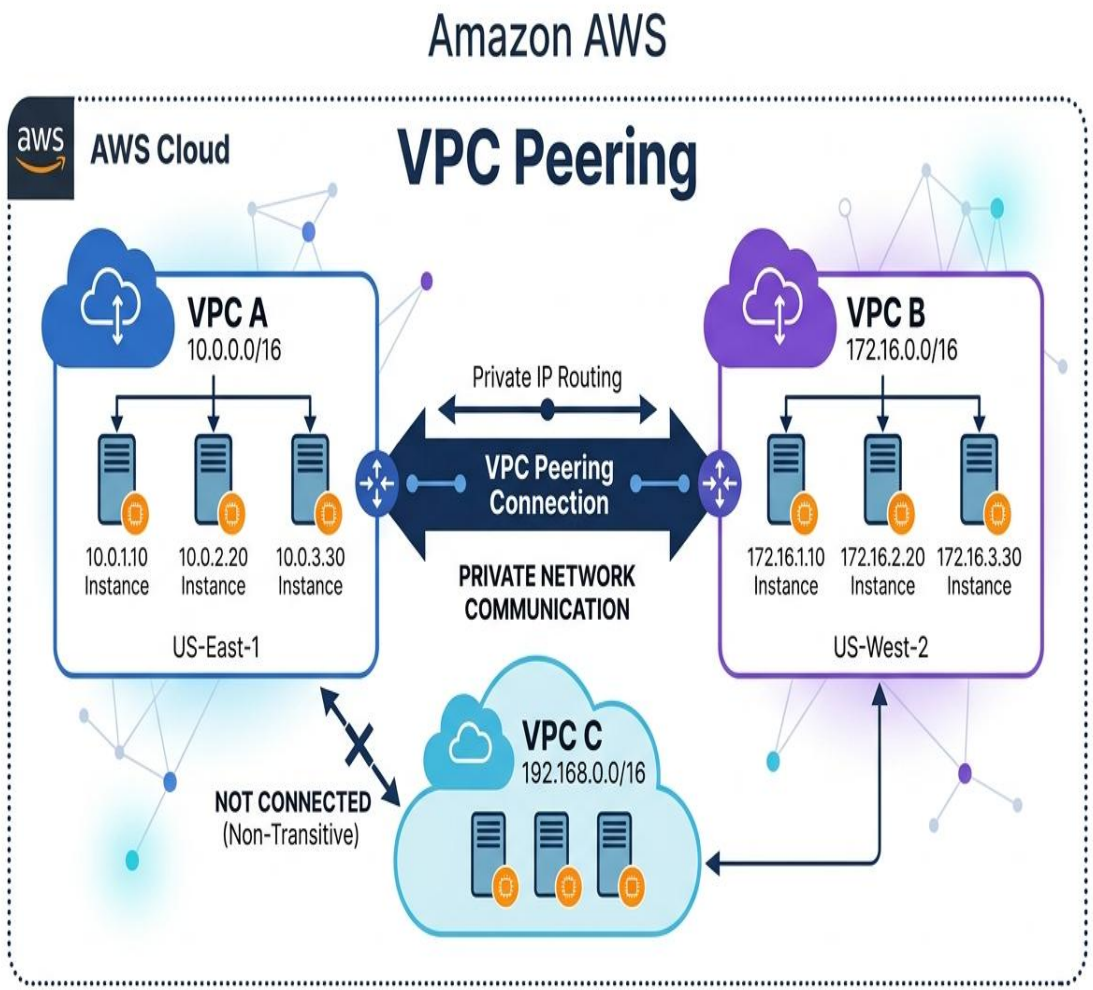
Production-ready reference architecture



- VPC spans all AZs in a Region
- Public + Private subnets in each AZ
- Load Balancer distributes across public subnets
- Web tier in public subnets (auto-scaled)
- App + DB tier in private subnets
- NAT Gateways per AZ for HA outbound traffic
- VPC Peering to connect with other VPCs
- This pattern is the AWS Well-Architected baseline

VPC Peering & Advanced Connectivity

Connecting multiple VPCs



- VPC Peering: Private connection between two VPCs (same or cross-account)
- Traffic stays on AWS backbone — no internet, no VPN
- Non-transitive: A↔B and B↔C does NOT mean A↔C
- CIDR blocks must NOT overlap
- Transit Gateway: Hub-and-spoke for many VPCs (scales better than peering mesh)
- VPN Gateway: Connects VPC to on-premises data centre
- AWS Direct Connect: Dedicated private line to AWS

Hands-On Lab — Build a 3-Tier VPC

Apply what you've learned in 20 minutes

1 Create VPC

CIDR: 10.0.0.0/16
Name: lab-vpc

2 Public Subnet

CIDR: 10.0.1.0/24
AZ: us-east-1a

3 Private Subnet

CIDR: 10.0.2.0/24
AZ: us-east-1a

4 Internet Gateway

Create & attach to lab-vpc

5 NAT Gateway

Place in public subnet
Allocate Elastic IP

6 Route Tables

Public: 0.0.0.0/0 → IGW
Private: 0.0.0.0/0 → NAT

🎯 *Verify: Launch EC2 in public subnet → ping 8.8.8.8. Launch EC2 in private subnet → should reach internet only via NAT.*

Key Takeaways & Summary

What you should know after today's session

🔵 VPC / VNet

A logically isolated private network in the cloud. You control IP ranges, subnets, routing and security.

🟢 Subnets & CIDR

Subnets divide the VPC into segments. CIDR notation defines the IP block size. /16 = 65K IPs, /24 = 256 IPs.

🟠 Internet Gateway

Enables bidirectional internet access for PUBLIC subnets. One IGW per VPC, attached directly.

🟣 NAT Gateway

Enables OUTBOUND-only internet for PRIVATE subnets. Hides private IPs from the internet.

🔴 Route Tables

Control where traffic flows. Public subnets point to IGW; private subnets point to NAT.

⚪ Security Layers

Security Groups (stateful, instance-level) + NACLs (stateless, subnet-level) = defense in depth.

🚀 **Next Session: Cloud Storage — S3, Blob Storage, Buckets, Object vs Block vs File Storage**

Practice: Complete the hands-on VPC lab before the next class.